DESIGN

Portfolio

– Ananya R

Project 1

To design a product based on the needs and pain points of the selected target audience.



To redesign a bed table for people who use laptops and writing tools, keeping in consideration hand (wrist) and neck strain experienced while using the existing designs in the market.

01. Primary Research : In this type of study, we survey or interview people to learn about their problems and challenges in the relevant field.

02. Secondary Research : In this type of study, we use pre-existing information sources such as the internet, research papers, blogs, journals, YouTube videos, books, magazines and so on to learn more about the subject.

03. Following an analysis of the primary and secondary research, we select a particular user group and attempt to employ user personas to concentrate on their specific requirements and pain issues.

Research

Research Outcomes :

Primary Research :

- The research highlights that homemakers' lifestyles are dynamic and strongly influenced by their children's life stage.
- While younger homemakers prioritize family and structured routines, older homemakers shift focus towards personal health, stability and social engagement.
- Across both, the core needs are balance, support and inspiration to adapt to changing responsibilities.

Secondary Research :

- The concept of lifestyle extends beyond daily habits and reflects individual identity, long-term choices, and cultural influences. Lifestyle is shaped by eating, clothing, living, transport, leisure, and communication patterns, all of which tie into sustainability debates.
- Contemporary research shows that future lifestyles will increasingly prioritize eco-friendly practices, service-oriented consumption, and sustainable growth. This means products must not only serve functional needs but also align with values of resource efficiency, reusability, and long-term well-being.
- For design, this highlights the need to integrate sustainability seamlessly into everyday choices, ensuring products support identity, convenience, and environmental responsibility.

Problem statement :

Students and corporate workers often rely on bed tables for studying or working long hours in confined or flexible spaces. However, most existing designs are bulky, lack ergonomic support, and do not adequately address posture, wrist strain, or multifunctional needs. There is a need for a lightweight, portable, and ergonomic bed table that supports comfort, productivity, and adaptability across diverse work and study environments.

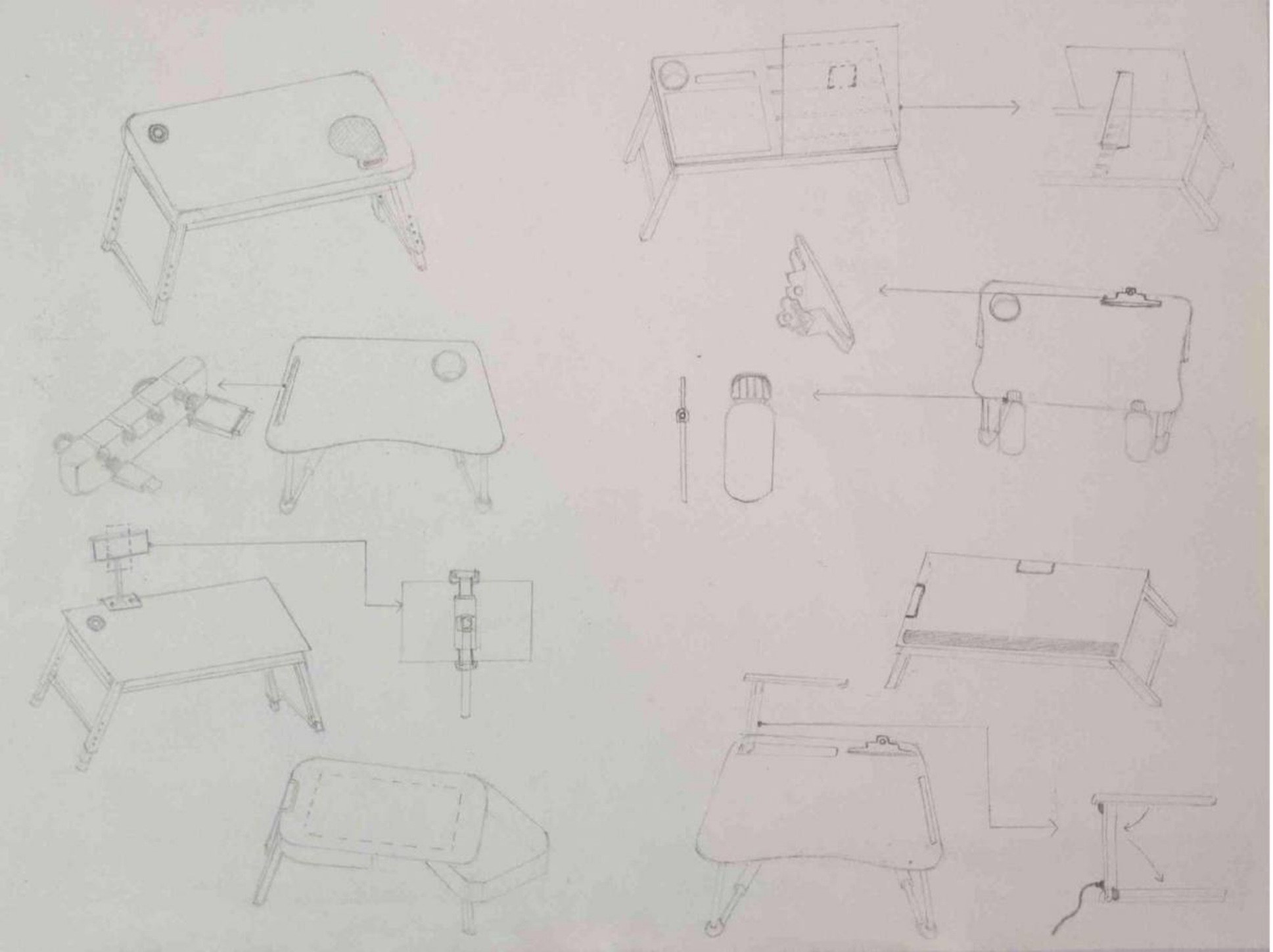
Redesign a multipurpose bed table that :

- Provides inclination and ergonomic positioning to reduce posture and wrist strain.
- Integrates modular features like phone holder, slot for charging cable, cup/bottle holder, and anti-slip feature for versatile use.
- Includes storage solutions such as slide-open compartments for stationery or documents.
- Folds compactly with a briefcase-style handle for portability and easy storage in smaller living/work spaces.

Problem Statement

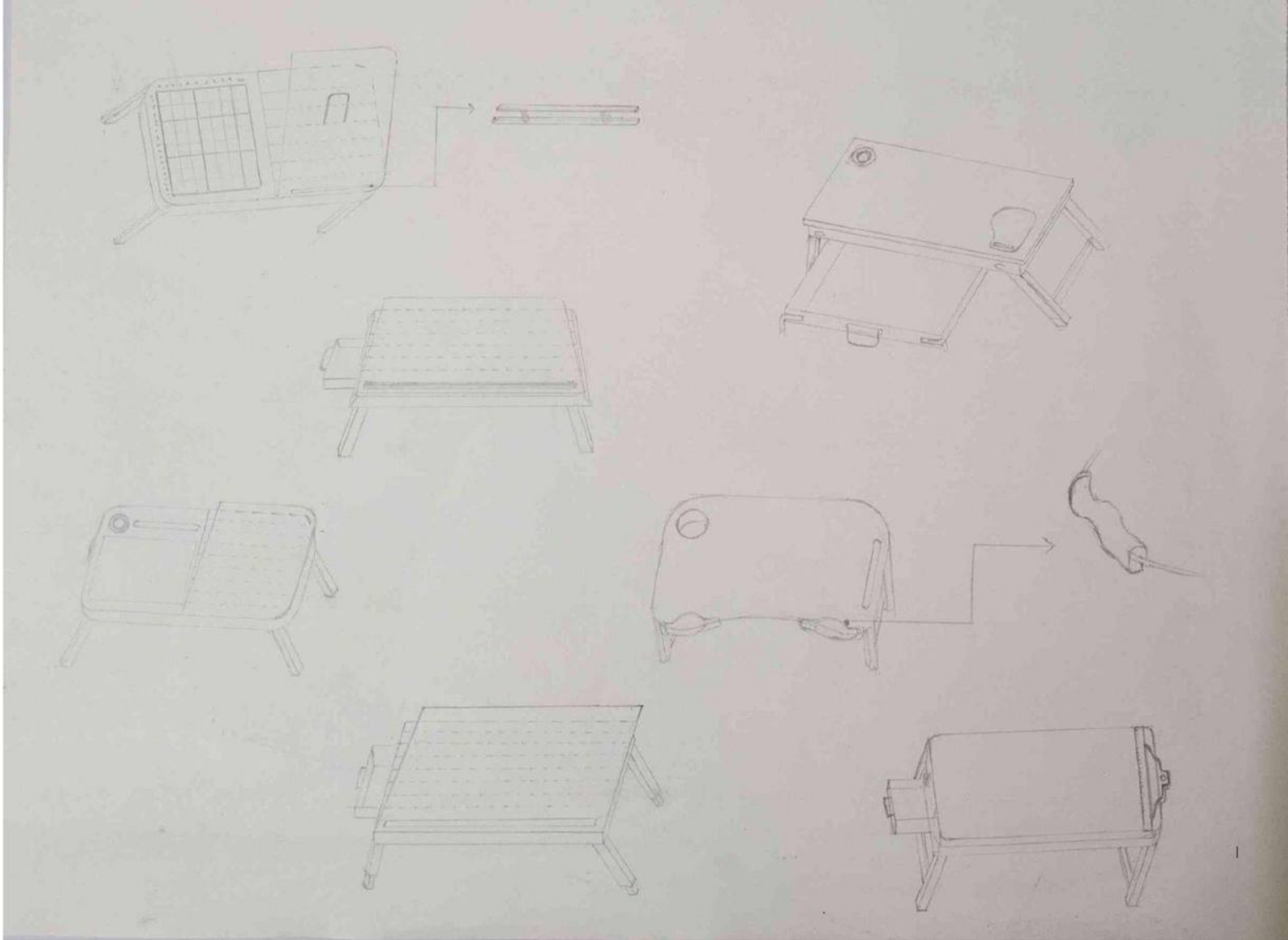
Companies →	Ikea	AmazonBasics	Wakefit	Pepperfry	Portronics	Bluewud
Portable	Yes	Yes	Yes	No	Yes	Yes
Foldable	Yes	Yes	No	No	Yes	Yes
Weight	1.2kgs	2kgs	19.5kgs	38kgs	2.65kgs	5.5kgs
Storage	No	Yes	No	No	No	No
Height Adjustment	No	No	No	Yes	Yes	No
Price	Rs.1,290	Rs. 649	Rs. 5,369	Rs. 7,999	Rs. 3,349	Rs. 2,399
Inclination	No	No	No	No	Yes	No
Form & Aesthetics	Basic	Good	Basic	Basic	Good	Basic
Sustainability	Yes	Partially	Yes	Partially	Partially	Yes

Market Research



Ideation Sketches

Ideation Sketches



Pugh chart of ideations

Features	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Portable	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Foldable	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Storage	N	N	N	N	N	N	Y	N	N	Y	Y	N	N	Y	Y
Height Adjustment	Y	N	N	N	Y	N	Y	N	N	Y	N	N	Y	Y	N
Inclination	N	Y	N	N	N	N	N	N	Y	N	Y	Y	N	Y	Y

BASE LENGTH - 13 cm
BASE BREADTH - 7 cm

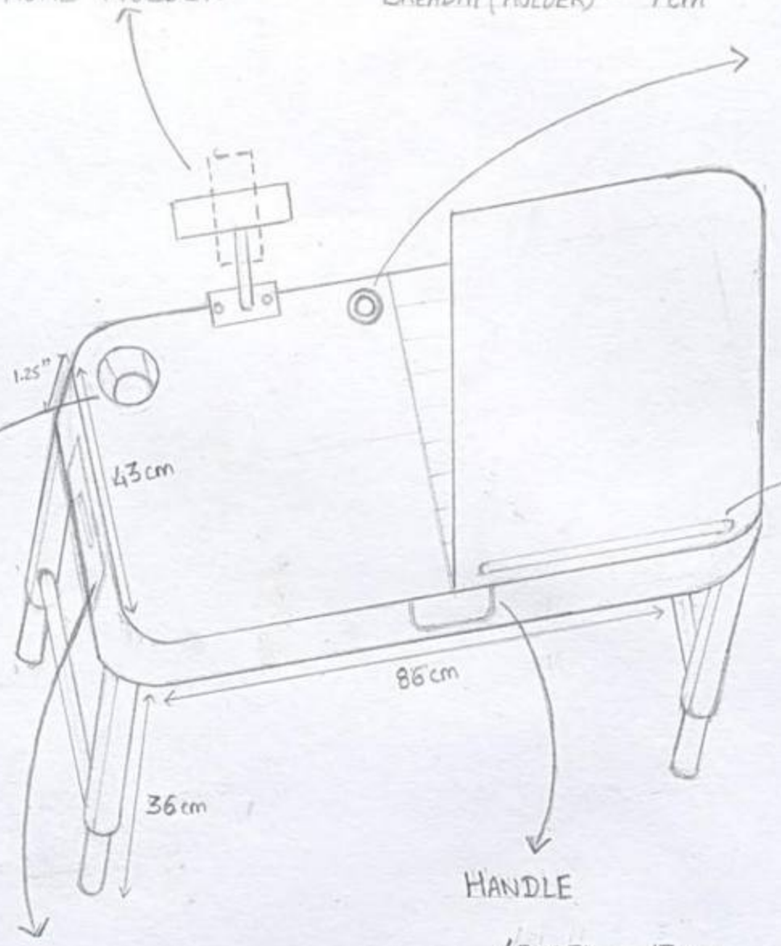
LENGTH (PHONE HOLDER ROD) - 15 cm
THICKNESS (PHONE HOLDER ROD) - 5 cm

LENGTH (HOLDER) - 15 cm
BREADTH (HOLDER) - 7 cm

PHONE HOLDER

CHARGER SLOT

OUTER CIRCUMFERENCE - 17 cm
INNER CIRCUMFERENCE - 15 cm



CUP HOLDER

CIRCUMFERENCE - 30.5 cm
DEPTH - 2.5 cm

SPRING GRIP

LENGTH - 38 cm
BREADTH - 2.5 cm
HEIGHT - 1.5 cm

HANDLE

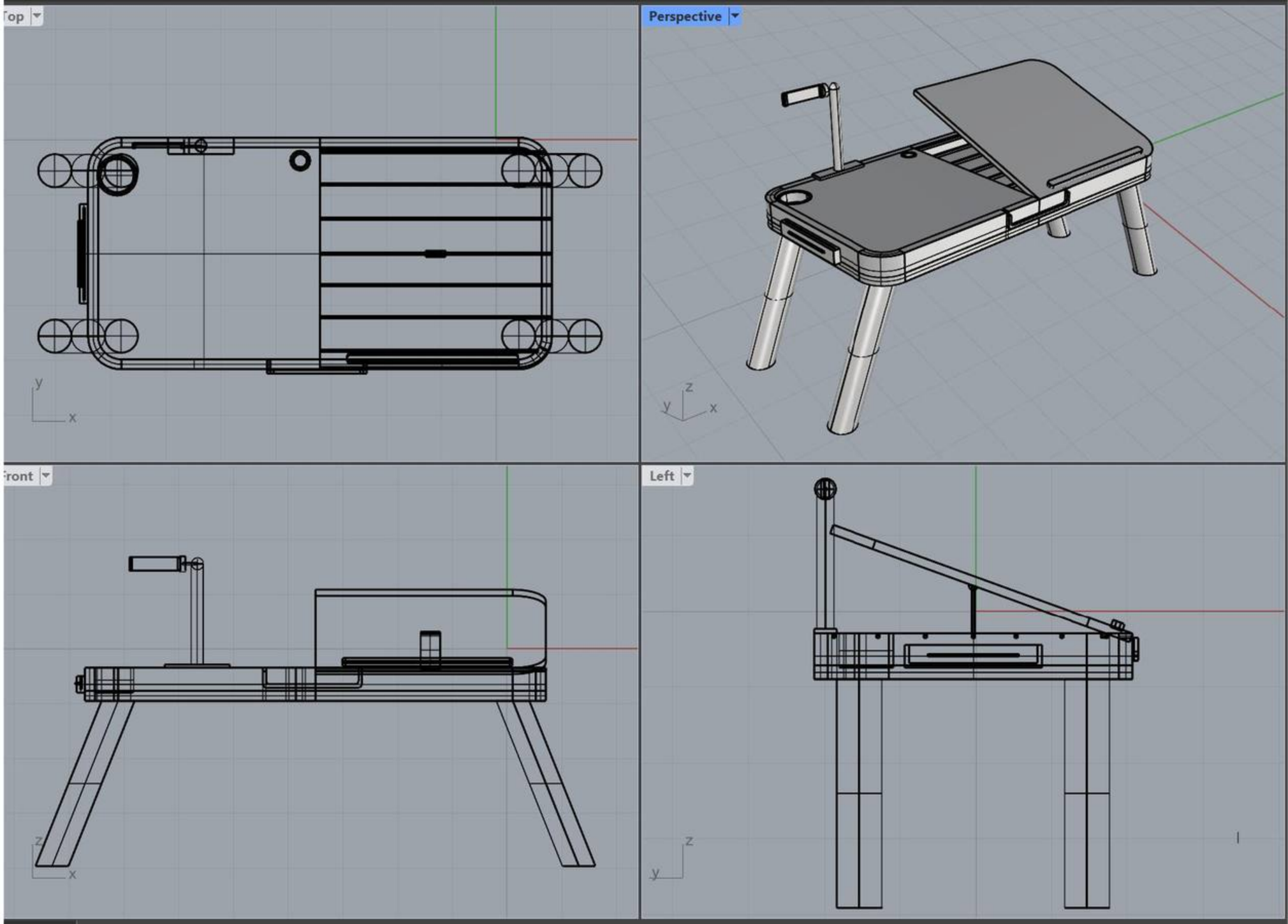
LENGTH - 13 cm
THICKNESS - 1.5 cm

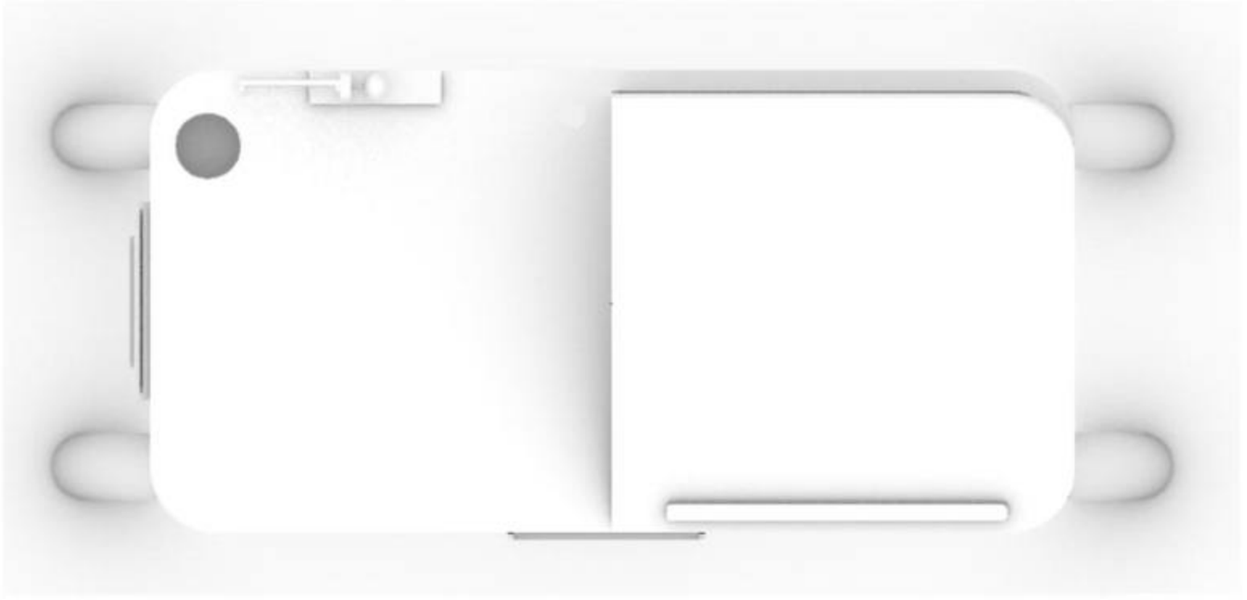
STORAGE TUB

LENGTH - 28 cm
BREADTH - 38 cm
HEIGHT - 2 cm

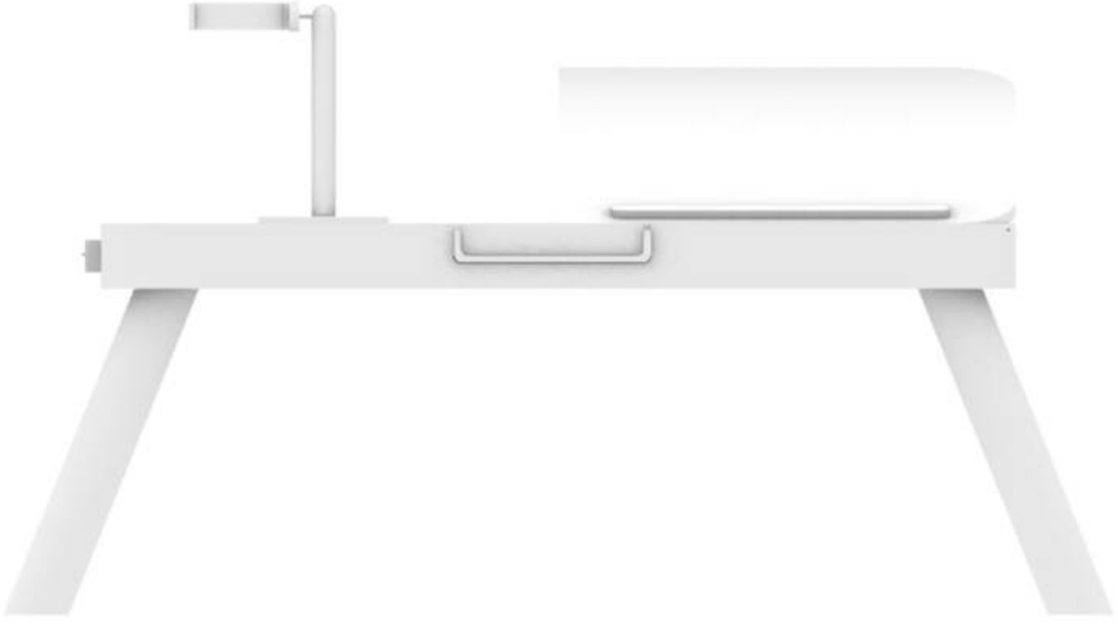
Final Sketch

3D Model





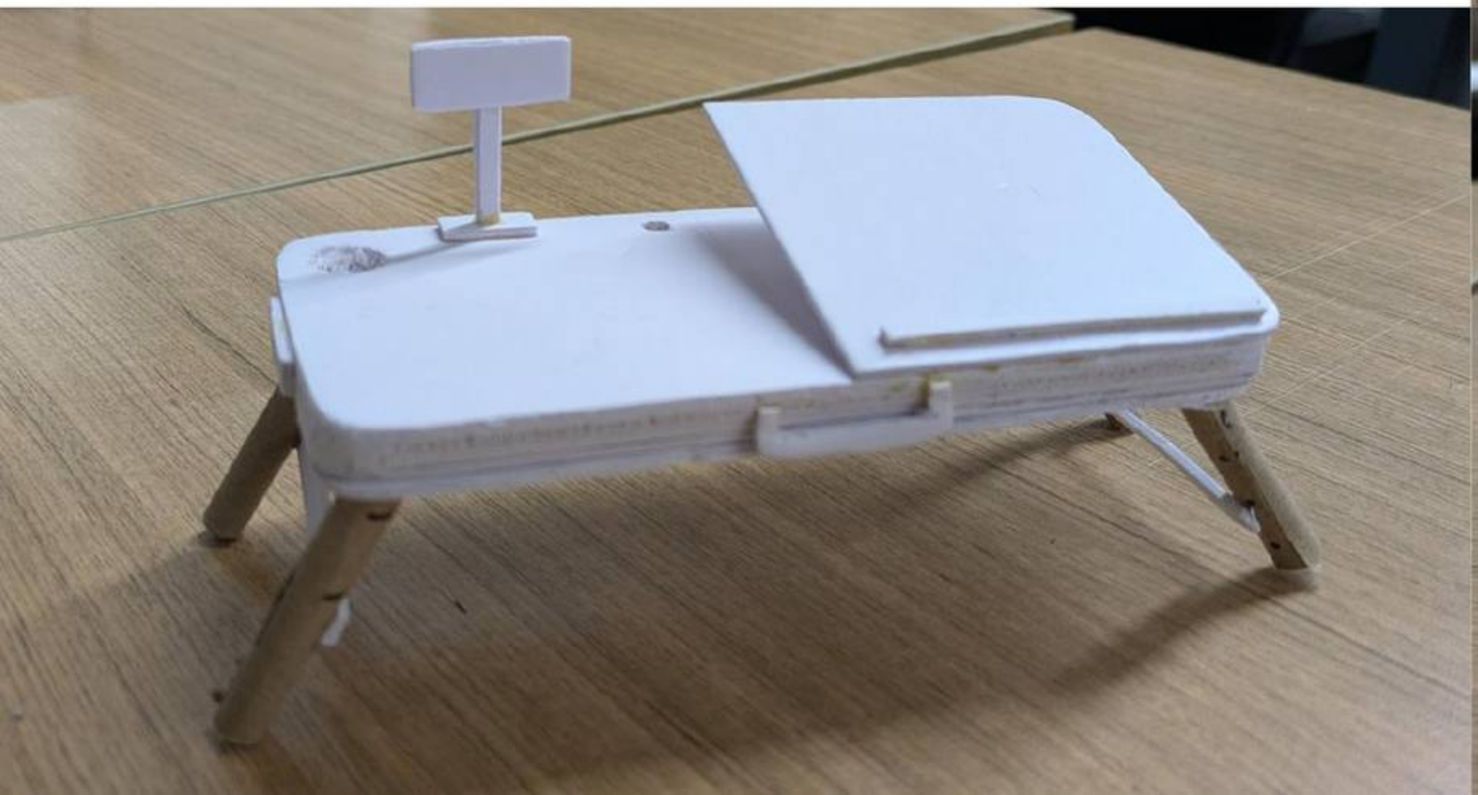






Mock - Up







Project 2

Product Analysis

Hair Curling Rod



About Product

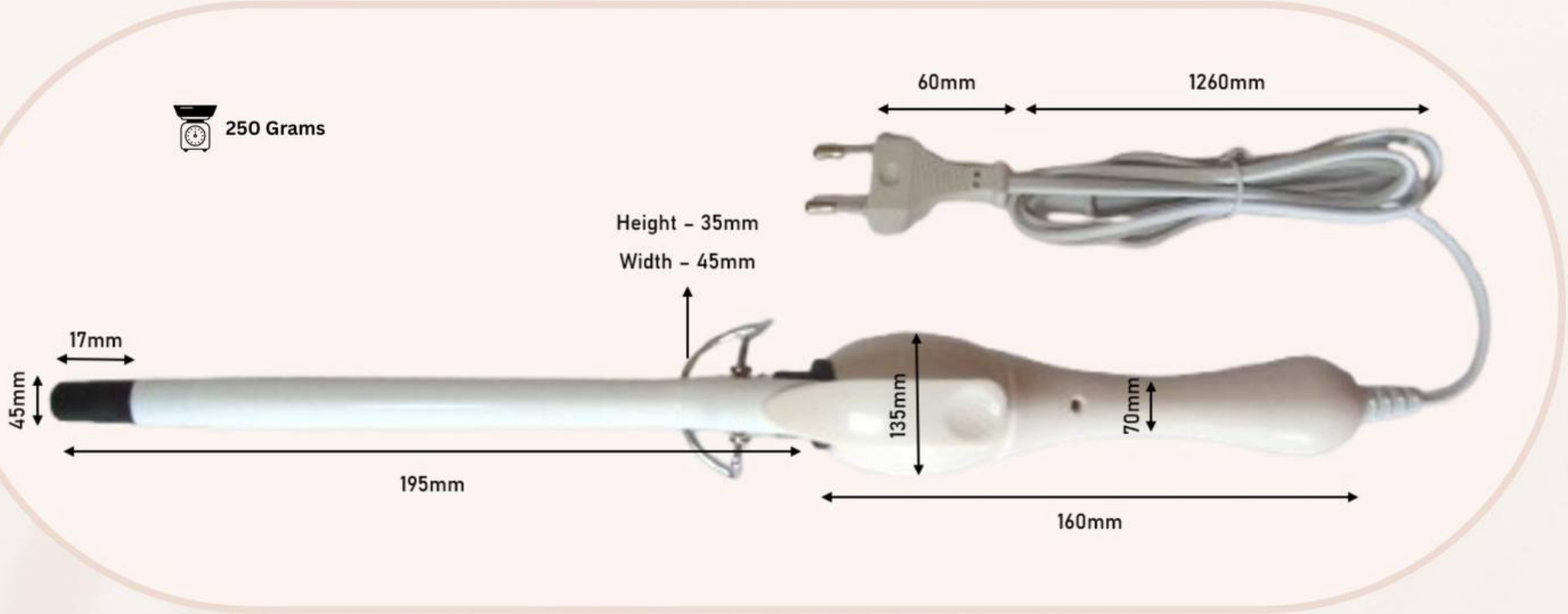
CONCEPTA

Brand - Concepta Hair Curler Machine

Cost - Rs. 1499

Features - The hair curler features a heated cylindrical barrel with a clamp to hold hair, along with an on/off switch, heat indicator light, insulated handle and a swivel cord for safe, convenient styling.

Dimensions


250 Grams
60mm
1260mm
Height – 35mm
Width – 45mm
17mm
45mm
195mm
135mm
70mm
160mm

Function

This product is used to curl and style hair by wrapping sections around a heated barrel.



1.

Turn on the curler and allow it to heat up.



2.

Place a section of hair and secure it with the clamp.



3.

Wrap the hair around the heated barrel and hold for a few seconds.



4.

Release the hair from the clamp and repeat this process.

CMF

Material

Colour

Off-white



Black

Ceramic-coated metal



Stanless steel



Molded ABS plastic

Finish



Semi-gloss plastic finish



Metallic finish



Slight reflective metal finish



Copper heating coil



PVC-coated insulated copper cable

Product Appraisal

• The Concepta Hair Curler is highly accessible because of its **affordable price**, making home styling possible for students and first-time users without financial commitment.

• Its simple clamp mechanism makes it **easy to handle** for beginners who may feel nervous using heated tools. The simple on/off switch keeps the experience **uncomplicated**.

• Visually the off-white body gives it a clean and minimal look. The design is **simple and functional** rather than overly styled or premium.

Product Concerns

1. **No temperature controls** to change as per individual's preference increasing chances of heat damage.
2. **Risk of burns** due to exposed metal barrel and less gap between clamp and barrel.
3. **No automatic shut off** compromising safety of users.
4. The clamp that holds the hair is not ergonomically comfortable and **demands a lot of physical strain** on the thumb.
5. **Not suitable for** users with **short hair** (shoulder length or shorter).
6. The product as such is **not sustainable** / environment friendly in any way.

Physical V/S Cognitive Ergonomic Chart

Product Name	Product Description	Feature Description	Physical / Cognitive	Positive / Negative	Severity
Hair Curling Iron	This product is used to create curls by wrapping hair around a heated cylindrical barrel.	1. No temperature control for different hair types.	Physical	Negative	High
	User needs to plug in the product and turn it on to start heating in set temperature.	2. Minimal white design gives aesthetically pleasing appearance.	Cognitive	Positive	Low
	The hair is wrapped around the barrel once it is heated to the temperate required.	3. Exposed hot barrel increases burn risk.	Physical	Negative	High
	After a few seconds clamp is pressed to loosen hair to remove it from barrel.	4. Barrel size and clamp design make it less suitable for styling short hair.	Physical	Negative	Low
	User is left with clean curls as hair takes the shape of the heated barrel.	5. Safety compromised due to no automatic shut-off feature.	Cognitive	Negative	Medium
		6. Lightweight body makes the curler easy to hold and style.	Physical	Positive	Medium
		7. Clamp requires significant thumb pressure causing strain.	Physical	Negative	Medium

Rating Parameters		
Severity	In Case of Negative	In Case of Positive
High	When the function fails or creates unsafe / hazardous situations for human.	Without the feature, it would be impossible to fulfill the task function.
Medium	It creates difficulty / additional efforts to achieve the task.	Enhances the task.
Low	A small deterent / improvement area observed.	An additional delight / comfort.

Score	

Score Rational :
7 to 10 is Very Good, 1 to 6 is Good, 0 to – 3 is satisfactory, – 4 to – 10 is very bad and needs immediate improvement.

-10 -9 -8 -7 -6 -5 -4 -3 -2 -1 0 1 2 3 4 5 6 7 8 9 10

Suggestions

• Introducing **adjustable temperature** settings or a **digital heat control system** with multiple heat levels.

• Adding a **longer insulated cool tip**, increasing the clamp – barrel gap and providing a silicone heat guard or protective hand gloves.

• Integrating an **automatic shut-off system** that turns the device off after inactivity or once the product has heated up to the maximum capacity.

• Redesigning the clamp with an **ergonomic thumb rest**, softer grip material and reduced spring tension.

• Introducing **interchangeable barrels** with smaller diameters.

• Using recyclable or bio-based plastics and designing the product for **easier repair and disassembly.**



Redesigned Product



Adjustable temperature of 120, 160 & 180 degrees celcius.

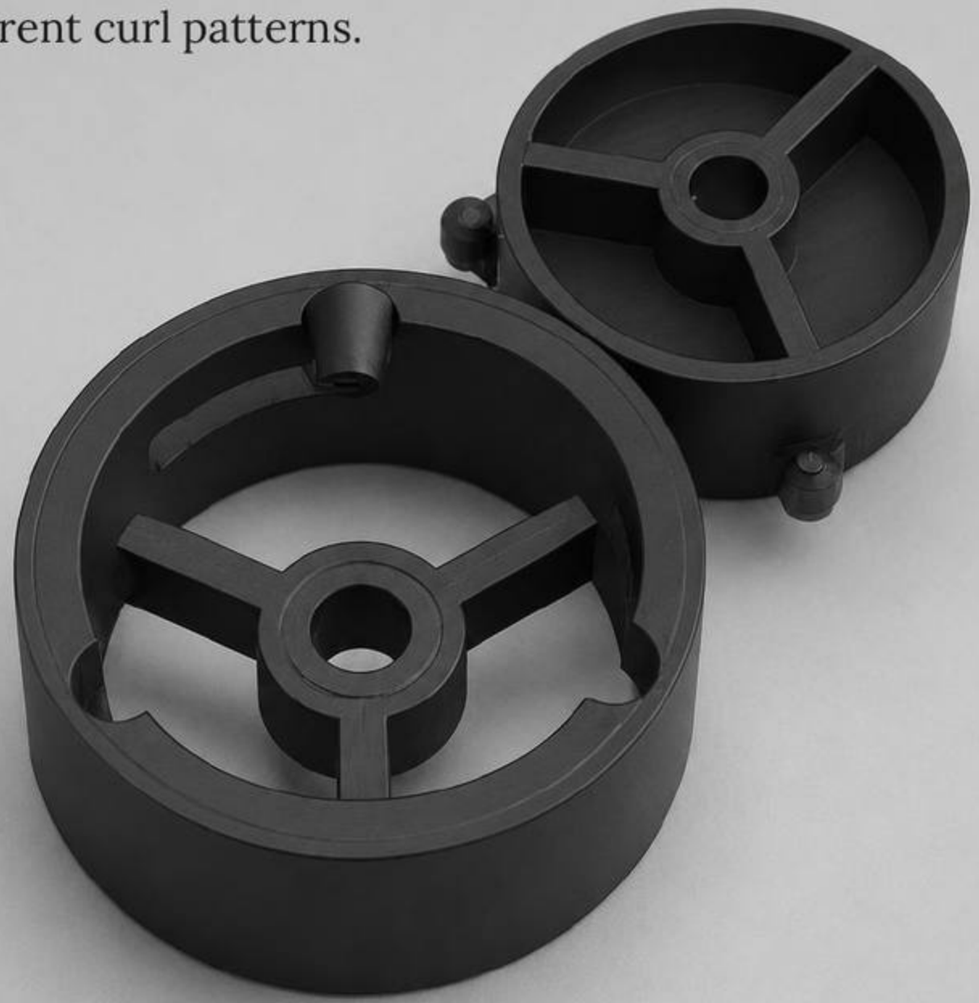


Clear protective case for safety and easier storage.





3 interchangeable barrels
for different curl patterns.



Secure twist-Lock mechanism
for easy barrel replacement.





Rechargeable via USB-C with LED status indicator.



Integrated touch sensor for Automatic rotation of spira barrel.

Renders


N°5
PREMIUM
SOLUTION



Human Interaction

Ai Generated


EFFORTLESS
CURLS.
ZERO EFFORT.







Thank You !